

41.4 The EM algorithm

Maximizing the observed-data likelihood Eq. 41.3.2 directly means differentiating an integral. EM avoids that: it alternates between filling in the missing pieces of the *complete-data log-likelihood* and maximizing the filled-in version, and it never decreases the observed-data likelihood. The iteration of Theorem 18.6.1(c) — fill each lost plot with its fitted value, refit, repeat — is the oldest instance in this book.

41.4.1 The two steps

Let $\ell(\boldsymbol{\theta}) = \log f(\mathbf{D}_{\text{obs}} \mid \boldsymbol{\theta})$ and $\ell_c(\boldsymbol{\theta}) = \log f(\mathbf{D} \mid \boldsymbol{\theta})$ be the observed-data and complete-data log-likelihoods, the second a function of data we do not have. The mechanism is ignorable throughout, so Theorem 41.3.1 licenses working with ℓ .

Definition 41.4.1 — EM

Given a current value $\boldsymbol{\theta}^{(s)}$, define

$$Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(s)}) = E\{\ell_c(\boldsymbol{\theta}) \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\theta}^{(s)}\}, \quad (41.4.1)$$

the expectation being over $\mathbf{D}_{\text{mis}}$ under the conditional density $f(\mathbf{D}_{\text{mis}} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\theta}^{(s)})$. The **E step** computes $Q(\cdot \mid \boldsymbol{\theta}^{(s)})$; the **M step** sets $\boldsymbol{\theta}^{(s+1)}$ to a maximizer of it. If the M step only *increases* Q , the scheme is called **generalized EM**.

The E step is not “fill in the missing values”; it fills in whatever functions of them the complete-data log-likelihood contains. When ℓ_c is linear in a set of sufficient statistics, as for every exponential family (Definition 34.1.1), the E step reduces to filling in those statistics ([Exercise \(B2\)](#)) — which is the same as filling in the missing values only when the statistics are linear in them. A normal variance, whose statistic contains $\sum_i y_i^2$, is the standing counterexample, and only the second reading is EM.

41.4.2 Monotonicity

Theorem 41.4.1 — The EM sequence never decreases the likelihood

Let $\boldsymbol{\theta}^{(s+1)}$ satisfy $Q(\boldsymbol{\theta}^{(s+1)} \mid \boldsymbol{\theta}^{(s)}) \geq Q(\boldsymbol{\theta}^{(s)} \mid \boldsymbol{\theta}^{(s)})$. Then

- (a) $\ell(\boldsymbol{\theta}^{(s+1)}) \geq \ell(\boldsymbol{\theta}^{(s)})$, with equality only if $Q(\boldsymbol{\theta}^{(s+1)} \mid \boldsymbol{\theta}^{(s)}) = Q(\boldsymbol{\theta}^{(s)} \mid \boldsymbol{\theta}^{(s)})$ and the two conditional densities of $\mathbf{D}_{\text{mis}}$ agree almost everywhere;
- (b) if ℓ is bounded above, $\ell(\boldsymbol{\theta}^{(s)})$ converges;
- (c) if $Q(\cdot \mid \cdot)$ is differentiable in its first argument and $\boldsymbol{\theta}^{(s+1)}$ is an interior maximizer, then at any fixed point $\boldsymbol{\theta}^*$ of the iteration the observed-data score vanishes: $\partial \ell(\boldsymbol{\theta}) / \partial \boldsymbol{\theta} \big|_{\boldsymbol{\theta}^*} = \mathbf{0}$.

Proof. Write $g(\mathbf{D}_{\text{mis}} | \boldsymbol{\theta}) = f(\mathbf{D}_{\text{mis}} | \mathbf{D}_{\text{obs}}, \boldsymbol{\theta})$ for the conditional density of the missing part. Since $f(\mathbf{D} | \boldsymbol{\theta}) = f(\mathbf{D}_{\text{obs}} | \boldsymbol{\theta}) g(\mathbf{D}_{\text{mis}} | \boldsymbol{\theta})$, taking logarithms and rearranging gives, for every $\mathbf{D}_{\text{mis}}$,

$$\ell(\boldsymbol{\theta}) = \ell_c(\boldsymbol{\theta}) - \log g(\mathbf{D}_{\text{mis}} | \boldsymbol{\theta}).$$

The left side does not depend on $\mathbf{D}_{\text{mis}}$, so taking the expectation of both sides under $g(\cdot | \boldsymbol{\theta}^{(s)})$ leaves it unchanged and yields the decomposition

$$\begin{aligned} \ell(\boldsymbol{\theta}) &= Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(s)}) - H(\boldsymbol{\theta} | \boldsymbol{\theta}^{(s)}), \quad \text{where} \\ H(\boldsymbol{\theta} | \boldsymbol{\theta}^{(s)}) &= \mathbb{E}\{\log g(\mathbf{D}_{\text{mis}} | \boldsymbol{\theta}) | \mathbf{D}_{\text{obs}}, \boldsymbol{\theta}^{(s)}\}. \end{aligned} \tag{41.4.2}$$

(a) Subtract Eq. 41.4.2 at $\boldsymbol{\theta}^{(s)}$ from the same at $\boldsymbol{\theta}^{(s+1)}$:

$$\begin{aligned} \ell(\boldsymbol{\theta}^{(s+1)}) - \ell(\boldsymbol{\theta}^{(s)}) &= \{Q(\boldsymbol{\theta}^{(s+1)} | \boldsymbol{\theta}^{(s)}) - Q(\boldsymbol{\theta}^{(s)} | \boldsymbol{\theta}^{(s)})\} \\ &\quad - \{H(\boldsymbol{\theta}^{(s+1)} | \boldsymbol{\theta}^{(s)}) - H(\boldsymbol{\theta}^{(s)} | \boldsymbol{\theta}^{(s)})\}. \end{aligned}$$

The first brace is nonnegative by hypothesis. For the second, write $g_s = g(\cdot | \boldsymbol{\theta}^{(s)})$ and $g_{s+1} = g(\cdot | \boldsymbol{\theta}^{(s+1)})$; then

$$\begin{aligned} H(\boldsymbol{\theta}^{(s+1)} | \boldsymbol{\theta}^{(s)}) - H(\boldsymbol{\theta}^{(s)} | \boldsymbol{\theta}^{(s)}) &= \int g_s \log \frac{g_{s+1}}{g_s} \\ &\leq \log \int g_s \frac{g_{s+1}}{g_s} = \log 1 = 0, \end{aligned}$$

by Jensen's inequality applied to the concave logarithm. So the difference of log-likelihoods is a nonnegative number minus a nonpositive one. Equality throughout forces equality in Jensen's inequality, which for the strictly concave logarithm requires g_{s+1}/g_s to be constant almost everywhere under g_s , hence $g_{s+1} = g_s$, and forces the first brace to vanish.

(b) A nondecreasing sequence bounded above converges.

(c) At a fixed point, $\boldsymbol{\theta}^*$ maximizes $Q(\cdot | \boldsymbol{\theta}^*)$, so $\partial Q(\boldsymbol{\theta} | \boldsymbol{\theta}^*)/\partial \boldsymbol{\theta}$ vanishes at $\boldsymbol{\theta} = \boldsymbol{\theta}^*$. The function $\boldsymbol{\theta} \mapsto H(\boldsymbol{\theta} | \boldsymbol{\theta}^*)$ is maximized at $\boldsymbol{\theta} = \boldsymbol{\theta}^*$ by the Jensen argument just given, so its derivative vanishes there too. Differentiating Eq. 41.4.2 at $\boldsymbol{\theta}^*$ gives $\partial \ell / \partial \boldsymbol{\theta} = \mathbf{0} - \mathbf{0}$. ■

Warning. Part (c) says fixed points are stationary points, not maxima: EM can converge to a saddle point or a local maximum, and with several modes the limit depends on the start. Wu (1983) gives conditions under which the limit points are stationary and under which the sequence converges; none rules out a local maximum. Start from several values.

41.4.3 Missing responses in a linear model

The oldest case is the one where every step is explicit, and it settles a question left open in Section 18.6.

Proposition 41.4.2 — EM for a normal linear model with missing responses

Let $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ and $\text{rank}(\mathbf{X}) = p$; let the responses of a set of m rows be unrecorded, the covariates of those rows being known, with an ignorable mechanism, and suppose the observed rows $\mathbf{X}_o$ have rank p . Write SSE_o for the residual sum of squares of the least squares fit on the observed rows. Then:

- (a) the E step replaces each missing y_i by $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^{(s)}$ in the cross-product terms of ℓ_c , and each y_i^2 by $(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^{(s)})^2 + \sigma^{2(s)}$;
- (b) the M step is $\boldsymbol{\beta}^{(s+1)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}^{(s)}$ with $\mathbf{y}^{(s)}$ the completed vector, and $\sigma^{2(s+1)} = n^{-1} \{ \|\mathbf{y}^{(s)} - \mathbf{X}\boldsymbol{\beta}^{(s+1)}\|^2 + m\sigma^{2(s)} \}$;
- (c) the unique fixed point is $\boldsymbol{\beta}^* = \hat{\boldsymbol{\beta}}_o$, the least squares estimate from the observed rows, and

$$\sigma^{2*} = \frac{\text{SSE}_o}{n - m}, \quad (41.4.3)$$

which is the observed-data maximum likelihood estimate, based on $n - m$ observations.

Proof.

- (a) The complete-data log-likelihood $-\frac{1}{2}n \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i (y_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta})^2$ is quadratic in the y_i , so its conditional expectation needs their first two conditional moments. Given the observed data an unrecorded y_i is a draw from its own model, so those are $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^{(s)}$ and $(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^{(s)})^2 + \sigma^{2(s)}$.
- (b) With those substitutions,

$$Q = -\frac{1}{2}n \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \left\{ \|\mathbf{y}^{(s)} - \mathbf{X}\boldsymbol{\beta}\|^2 + m\sigma^{2(s)} \right\},$$

because each missing row contributes $E(y_i^2) - 2\mathbf{x}_{(i)}^\top \boldsymbol{\beta} E(y_i) + (\mathbf{x}_{(i)}^\top \boldsymbol{\beta})^2 = (\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^{(s)})^2 + \sigma^{2(s)}$. Maximizing in $\boldsymbol{\beta}$ is ordinary least squares on $\mathbf{y}^{(s)}$; maximizing in σ^2 gives the stated average.

- (c) At a fixed point the completed vector has $y_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}^*$ at the missing rows, so those rows have zero residual and contribute nothing to the least squares criterion; hence $\boldsymbol{\beta}^*$ minimizes $\|\mathbf{y}_o - \mathbf{X}_o \boldsymbol{\beta}\|^2$ and equals $\hat{\boldsymbol{\beta}}_o$, which is Theorem 18.6.1(a). Its residual sum of squares is SSE_o , so the variance equation reads $n\sigma^{2*} = \text{SSE}_o + m\sigma^{2*}$, giving Eq. 41.4.3. Uniqueness of $\boldsymbol{\beta}^*$ follows from Theorem 18.6.1(b), and Eq. 41.4.3 then determines σ^{2*} .

■

The term $m\sigma^{2(s)}$ is the whole point. Without it — fill the gaps with fitted values and hand the completed data to a regression routine — the variance comes out as SSE_o/n . That is the error Theorem 18.6.1(e) warns against, now derived rather than asserted: *the filled-in values are not data, and the likelihood knows it.*

Example. (The Mauna Loa gaps)

Fit the trend-plus-harmonics model of [Section 5.3](#) to all 2284 weeks, with the 59 missing ones handled by EM. The coefficients agree with the complete-case fit to machine precision, as [Proposition 41.4.2\(c\)](#) requires, and the error standard deviation is 0.7992 ppm; filling in the gaps and treating them as data gives 0.7889 ppm, too small by the factor $\sqrt{n/(n-m)} = 1.0132$. The discrepancy is small here only because the gaps are few.

```
import numpy as np
import statsmodels.api as sm

co2 = sm.datasets.co2.load_pandas().data["co2"]
t = (co2.index.year + (co2.index.dayofyear - 0.5) / 365.25).to_numpy()
tc = t - 1980.0
X = np.column_stack([np.ones(len(t)), tc, tc**2,
                     np.cos(2 * np.pi * t), np.sin(2 * np.pi * t),
                     np.cos(4 * np.pi * t), np.sin(4 * np.pi * t)])

y = co2.to_numpy()
obs = ~np.isnan(y)
n, m, p = len(y), int((~obs).sum()), X.shape[1]

beta, sigma2 = np.zeros(p), 1.0 # any starting value will do
for sweep in range(60):
    filled = np.where(obs, y, X @ beta) # E step: gaps get their
    fitted values
    beta, *_ = np.linalg.lstsq(X, filled, rcond=None) # M step for the coefficients
    sse = float(np.sum((filled - X @ beta) ** 2))
    sigma2 = (sse + m * sigma2) / n # M step for the variance: add
    m*sigma^2
print(f"EM after {sweep + 1} sweeps: sigma-hat = {np.sqrt(sigma2):.4f} ppm")

beta_cc, *_ = np.linalg.lstsq(X[obs], y[obs], rcond=None)
sse_o = float(np.sum((y[obs] - X[obs] @ beta_cc) ** 2))
print(f"observed-data MLE: {np.sqrt(sse_o / (n - m)):.4f} ppm")
print(f"treating the fills as data: {np.sqrt(sse_o / n):.4f} ppm")
```

[Run this code online](#)

41.4.4 A missing covariate

When a covariate is missing, the conditional distribution the E step needs runs *backwards*: what does a recorded response say about an unrecorded covariate? A model for the covariate is now unavoidable.

Proposition 41.4.3 — EM with a missing covariate

Let $x_i \sim N(\mu, \tau^2)$ independently, and $y_i | x_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$, with $\theta = (\beta_0, \beta_1, \sigma^2, \mu, \tau^2)$. Suppose y_i is always recorded and x_i is missing for a subset, with an ignorable mechanism. Then for a record with x_i missing the E step uses

$$E(x_i | y_i) = \mu + \frac{\beta_1 \tau^2}{\beta_1^2 \tau^2 + \sigma^2} (y_i - \beta_0 - \beta_1 \mu), \quad \text{Var}(x_i | y_i) = \tau^2 - \frac{\beta_1^2 \tau^4}{\beta_1^2 \tau^2 + \sigma^2}, \quad (41.4.4)$$

both evaluated at $\theta^{(s)}$, and the M step is the complete-data maximum likelihood fit with the sufficient statistics $\sum_i x_i$, $\sum_i x_i^2$ and $\sum_i x_i y_i$ replaced by $\sum_i E(x_i | y_i)$, $\sum_i E(x_i^2 | y_i)$ and $\sum_i y_i E(x_i | y_i)$.

Proof. The pair (x_i, y_i) is bivariate normal, with $E y_i = \beta_0 + \beta_1 \mu$, $\text{Var } y_i = \beta_1^2 \tau^2 + \sigma^2$ and $\text{Cov}(x_i, y_i) = \beta_1 \tau^2$. The conditional moments Eq. 41.4.4 are the standard normal conditional mean and variance (Theorem 3.5.2). The complete-data log-likelihood is

$$-\frac{n}{2} \log \sigma^2 - \frac{\sum_i (y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2} - \frac{n}{2} \log \tau^2 - \frac{\sum_i (x_i - \mu)^2}{2\tau^2} + \text{const},$$

which is linear in the three listed statistics together with $\sum_i y_i$ and $\sum_i y_i^2$; taking conditional expectations therefore replaces exactly those statistics, and the maximizer is the complete-data one evaluated at them. ■

Note what the E step does *not* do: it does not impute x_i by μ , nor from other covariates alone. It uses the response. Imputing a covariate without the response is among the commonest errors in practice, and it attenuates $\hat{\beta}_1$ towards zero, for the reason Chapter 24 gives.

Example. (An EM fit with a missing covariate)

Simulate $N = 400$ records with $\beta_1 = 0.8$, and let the covariate be missing more often when the response is large; this loses 169 covariate values. The mechanism is MAR — it uses only the recorded y — so likelihood is valid and the complete-case fit is not. EM gives $\hat{\beta}_1 = 0.825$; the complete cases give 0.620, attenuated by a quarter.

Figure 41.4.1 shows the two faces of the algorithm. The observed-data log-likelihood climbs monotonically, as Theorem 41.4.1(a) requires, and the distance to the fixed point falls by the constant factor 0.760 per sweep, matching to three decimals the largest eigenvalue of $\mathcal{I}_c^{-1} \mathcal{I}_m$ below. Reaching eight decimal places takes 68 sweeps — slow by the standards of Newton’s method, which is the price of never differentiating an integral.

```
rng = np.random.default_rng(4104)
N = 400
theta_true = dict(b0=1.0, b1=0.8, s2=1.0, mu=0.5, t2=1.0)
x = theta_true["mu"] + np.sqrt(theta_true["t2"]) * rng.normal(size=N)
y2 = theta_true["b0"] + theta_true["b1"] * x + np.sqrt(theta_true["s2"]) * rng.
    normal(size=N)
seen = rng.uniform(size=N) > 1 / (1 + np.exp(-(-0.4 + 2.0 * (y2 - y2.mean()))))
print(f"{N - seen.sum()} of {N} covariates missing, more often when y is large")

def estep(th, y, seen):
    """Conditional mean and second moment of the covariate given the response."""
    b0, b1, s2, mu, t2 = th
```

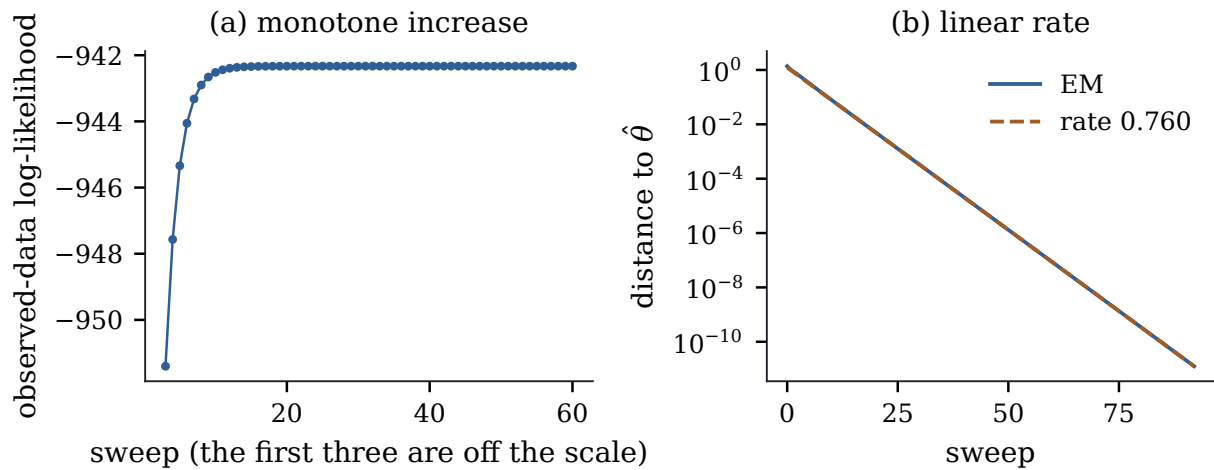


Figure 41.4.1: EM for a linear model with a missing covariate. (a) The observed-data log-likelihood increases at every sweep. (b) The distance from the current iterate to the fixed point decays geometrically at the rate predicted by the fraction of missing information.

```

vy = b1**2 * t2 + s2                                # marginal variance of y
cond_mean = mu + (b1 * t2 / vy) * (y - b0 - b1 * mu)
cond_var = t2 - (b1 * t2) ** 2 / vy
ex = np.where(seen, x, cond_mean)
ex2 = np.where(seen, x**2, cond_mean**2 + cond_var)
return ex, ex2

def mstep(ex, ex2, y):
    mu, n_ = ex.mean(), len(y)
    t2 = ex2.mean() - mu**2
    sxx = ex2.sum() - n_ * mu**2
    sxy = (ex * y).sum() - n_ * mu * y.mean()
    b1 = sxy / sxx
    b0 = y.mean() - b1 * mu
    s2 = ((y**2).sum() - 2 * b0 * y.sum() - 2 * b1 * (ex * y).sum()
          + n_ * b0**2 + 2 * b0 * b1 * ex.sum() + b1**2 * ex2.sum()) / n_
    return np.array([b0, b1, s2, mu, t2])

def loglik(th, y, seen):
    """Observed-data log-likelihood: joint for complete rows, marginal for the rest.
    """
    b0, b1, s2, mu, t2 = th
    ll = -0.5 * np.sum(np.log(2 * np.pi * s2) + (y[seen] - b0 - b1 * x[seen]) ** 2 /
                       s2)
    ll += -0.5 * np.sum(np.log(2 * np.pi * t2) + (x[seen] - mu) ** 2 / t2)
    vy = b1**2 * t2 + s2
    ll += -0.5 * np.sum(np.log(2 * np.pi * vy) + (y[~seen] - b0 - b1 * mu) ** 2 / vy
                       )
    return ll

th = np.array([0.0, 0.0, 1.0, 0.0, 1.0])
path, iterates = [loglik(th, y2, seen)], [th]
for sweep in range(200):
    ex, ex2 = estep(th, y2, seen)
    th = mstep(ex, ex2, y2)

```

```

path.append(loglik(th, y2, seen))
iterates.append(th)
print("EM   b0, b1 =", np.round(th[:2], 4))

cc = np.polyfit(x[seen], y2[seen], 1)[::-1]          # complete cases only
print("complete cases:", np.round(cc, 4))

```

[Run this code online](#)

41.4.5 Rate of convergence, and the standard errors EM does not give

Both the speed of EM and its principal defect come from the same matrix. Define the **complete-data information** $\mathcal{I}_c = -\partial^2 Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^*) / \partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top$ at $\boldsymbol{\theta} = \boldsymbol{\theta}^*$, the **observed information** $\mathcal{I}_o = -\partial^2 \ell / \partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top$ there, and their difference $\mathcal{I}_m = \mathcal{I}_c - \mathcal{I}_o$, the **missing information**.

Proposition 41.4.4 — The missing-information principle

At any $\boldsymbol{\theta}$, with $\mathbf{S}_c$ the complete-data score,

$$\frac{\partial \ell}{\partial \boldsymbol{\theta}} = \mathbb{E}(\mathbf{S}_c \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\theta}), \quad \mathcal{I}_o = \mathbb{E}(-\partial \mathbf{S}_c / \partial \boldsymbol{\theta}^\top \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\theta}) - \text{Var}(\mathbf{S}_c \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\theta}). \quad (41.4.5)$$

Consequently $\mathcal{I}_m = \text{Var}(\mathbf{S}_c \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\theta})$ is nonnegative definite: the observed information is never larger than the expected complete-data information.

Proof. Differentiate the identity $\ell(\boldsymbol{\theta}) = \ell_c(\boldsymbol{\theta}) - \log g(\mathbf{D}_{\text{mis}} \mid \boldsymbol{\theta})$ from the proof of [Theorem 41.4.1](#) and take the conditional expectation under $g(\cdot \mid \boldsymbol{\theta})$. The conditional score of g has mean zero, being the score of a density in its own parameter, so the first identity follows. Differentiating again,

$$-\frac{\partial^2 \ell}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top} = -\frac{\partial^2 \ell_c}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top} + \frac{\partial^2 \log g}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top},$$

and taking the conditional expectation turns the last term into minus the information of g , which by the usual information identity equals minus the variance of its score. Since the score of g is $\mathbf{S}_c - \partial \ell / \partial \boldsymbol{\theta}$ and the second term is a constant given the observed data, that variance is $\text{Var}(\mathbf{S}_c \mid \mathbf{D}_{\text{obs}})$. ■

Identity [Eq. 41.4.5](#) is Louis's formula (Louis 1982); “observed = complete minus missing” is the missing-information principle of Orchard and Woodbury (1972). Two consequences:

The rate. Near a fixed point the EM map is approximately linear with Jacobian $\mathbf{I} - \mathcal{I}_c^{-1} \mathcal{I}_o = \mathcal{I}_c^{-1} \mathcal{I}_m$, so the iterates converge linearly at its largest eigenvalue — the largest **fraction of missing information** (Dempster, Laird and Rubin, 1977). In [Example \(An EM fit with a missing covariate\)](#) that eigenvalue is 0.760 and the measured rate is 0.760; the log-likelihood

gap, quadratic near the maximum, falls at its square, 0.577 per sweep. In Theorem 18.6.1(c) the same matrix is $\mathbf{M}_{mm}$, the submatrix of the hat matrix belonging to the lost plots.

The standard errors. The matrix EM manipulates is $\mathcal{I}_c$, which is *too large*, so inverting it understates the variance by the fraction of missing information. Three honest routes: Eq. 41.4.5; numerical differentiation of ℓ , as this section’s script does; or the bootstrap, resampling records and rerunning EM (Definition 23.1.1(b)). In Example (An EM fit with a missing covariate) these give 0.0611 and 0.0605 (bootstrap, 300 resamples), against 0.0512 from the complete-data information — an understatement of 16%.

Warning. The variance of an EM estimate is never the inverse of the matrix the M step inverts. If a routine reports standard errors after EM without saying how it got them, assume it has made this mistake.

41.4.6 Exercises

41.4.6.1 A. Check your understanding

Exercise A1.

The EM algorithm for the zero-inflated model of Section 37.5 treats the component label as the missing datum. Identify $\mathbf{D}_{\text{mis}}$, Q and the M step there, and check that Exercise (B4) is the special case of Theorem 41.4.1(a).

41.4.6.2 B. Practice

Exercise B1.

In Proposition 41.4.2, suppose one carries out the iteration but *omits* the term $m\sigma^{2(s)}$ from the M step for the variance. Show that the coefficients are unaffected and that the variance converges to SSE_o/n . Relate the discrepancy to Theorem 18.6.1(e).

Solution. The coefficient update does not involve σ^2 at all, so $\hat{\boldsymbol{\beta}}^{(s)}$ follows the same sequence and converges to $\hat{\boldsymbol{\beta}}_o$; at that point the completed residual sum of squares is SSE_o . The truncated variance update is $\sigma^{2(s+1)} = \text{SSE}_o/n$ once the coefficients have settled. That is the complete-data formula applied to n “observations”, m of which have been manufactured to fit exactly; reducing the divisor by m is the degrees-of-freedom correction of Theorem 18.6.1(e).

Exercise B2.

Let the complete-data model be an exponential dispersion family (Definition 34.1.1) with sufficient statistic $\mathbf{t}(\mathbf{D})$. Show that the E step reduces to computing $E\{\mathbf{t}(\mathbf{D}) \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\theta}^{(s)}\}$ and the M step to solving the complete-data likelihood equations with $\mathbf{t}$ replaced by that expectation. Why does the normal variance in Proposition 41.4.2 still require the extra term $m\sigma^{2(s)}$?

Solution. The complete-data log-likelihood is $\boldsymbol{\theta}^\top \mathbf{t}(\mathbf{D}) - A(\boldsymbol{\theta})$ plus a term free of $\boldsymbol{\theta}$, so it is affine in $\mathbf{t}$ and its conditional expectation replaces $\mathbf{t}$ by $E(\mathbf{t} \mid \cdot)$. Setting the derivative

to zero gives $E_{\theta} \mathbf{t} = E(\mathbf{t} \mid \mathbf{D}_{\text{obs}})$. The normal case is no exception: the sufficient statistics include $\sum_i y_i^2$, and its conditional expectation over a missing row is $(\mathbf{x}_{(i)}^{\top} \boldsymbol{\beta}^{(s)})^2 + \sigma^{2(s)}$, which is where the extra term comes from. The trap is to substitute the conditional *mean of the data* rather than the conditional mean of the *statistic*.

41.4.6.3 C. Going deeper

Exercise C1.

Construct a two-component normal mixture with missing labels — the classical EM problem — for which the likelihood is unbounded, and show that EM started near a degenerate configuration drives one variance to zero while the likelihood increases at every step. Reconcile this with [Theorem 41.4.1\(b\)](#).